

MinKnow Set up

Below are the setup instructions for the different sequencing runs. **Please make sure to change the Experiment name and Sample ID to your own.**

1. Rapid Enrich - MinKnow

Starting a Sequencing Setup on GridION

1. Select the 'Start Sequencing' option on the Start homepage to choose the running parameters for your experiment.

Sequencing and analysis

**Start sequencing**
Configure and start a new sequencing run

**Start from a sample sheet**
Import sample sheet for sequencing

**Flow cell check**
Assess flow cell health

**Post sequencing analysis**
Analyse existing sequence data

**Custom scripts**
Start a run from a custom script

Utilities

**Hardware check**
Validate device is ready for sequencing

**Minimap2 converter**
Create a .mmi from a .fasta reference

Positions

2. Type in experiment name, select flow cell position, Add Sample ID, and choose flow cell type from the drop down menu.
 - **Experiment Name:** DSCWorkshop2025
 - **Position:** X1
 - **SampleID:** DSC2025_RAD_Enrich
 - Flow cell ID automatically is filled in
 - **Flow cell type:** FLO-MIN114

Positions

Experiment details

Experiment name *

DSCWorkshop2025

Device positions

MinION positions

GridION · INT-GRD-1577

[Deselect all positions](#)

▶

X1

X2

X3

X4

X5

Position	Flow cell ID	Flow cell type	Sample ID
X1	FAX56448	FLO-MIN114 ▾	DSC2025_RAD_Enrich

3. Select Continue to move to the Kit Selection.

4. Select the kit used from the Kit selection menu: **Rapid Sequencing Kit (SQK-RAD114)**

Kit selection

Sample type

DNA

RNA

PCR-free

PCR

PCR-free

Multiplexing

Yes

No

☐ Control

[Reset filters](#)

Ligation Sequencing Kit SQK-LSK114	Ligation Sequencing Kit SQK-LSK114-XL	Rapid Barcoding Kit 96 SQK-RBK114-96	Native Barcoding Sequencing Kit 24 SQK-NBD114-24
Native Barcoding Sequencing Kit 96 SQK-NBD114-96	Ultra-Long DNA Sequencing Kit SQK-ULK114	16S Barcoding Kit SQK-16S114-24	Multiplex Ligation Sequencing Kit XL SQK-MLK114-96-XL
PCR cDNA Barcoding Kit - 24 SQK-PCB114-24	PCR-cDNA Sequencing Kit SQK-PCS114	Rapid Sequencing Kit SQK-RAD114	Rapid Barcoding Kit 24 SQK-RBK114-24
Rapid PCR Barcoding Kit SQK-RPB114-24			

5. Select Continue to move to the Run configuration.

6. In Sequencing and analysis, basecalling, and Adaptive sampling, and advanced sequencing options can be modified.
 - a. Ensure Basecalling is selected (On)
 - b. Edit the model to select **Super-accurate basecalling** with Modified bases Off.
 - c. Ensure Barcoding is Off
 - d. Ensure Alignment is Off
 - e. Turn on Adaptive sampling
 - i. **Select Enrich**
 - ii. Ensure Basecall on-target reads only is Off
 - iii. **For Reference for alignment, choose the appropriate genome from the location of your reference genomes**
 - iv. Under advanced options, change the **End channel to 256**

Sequencing and analysis

Basecalling

Model

Super-accurate – v4.3.0 – 400bps

Modified bases

Off

Barcoding

Alignment

Adaptive sampling

Method

Enrich

Basecall on-target reads only

Off

Selected channels

1 – 256

Reference

Bacillus_subtilis_complete_genome.fasta

Bed file

No file selected

Advanced sequencing options

7. In Data targets, change the Run limit to Stop run when Time equals your desired run length (for the workshop we initially used 18 hrs).

Data targets

Conditions

Run limit

Stop run when sequencing reaches 18 hours

8. Output options can be left with defaults.
- a. File location: /data/
 - b. Data format: FASTQ
 - c. Output frequency: Every 10 minutes
 - d. FASTQ compression ON
 - e. Raw reads: POD5
 - f. Filtering On, Min Qscore = 10

Output

File output options

File location

/data/

346.2 GB used – 2.9 TB free

Data format

FASTQ

Output frequency

Every 10 minutes

FASTQ compression

On

Raw reads

POD5

Filtering

☒

Min Q score

10

9. Select 'Start' to run the experiment.

Rapid Deplete - MinKnow

Commented [1]: @kgrond@alaska.edu please review
Assigned to kgrond@alaska.edu

Starting a Sequencing Setup on GridION

1. Select the 'Start Sequencing' option on the Start homepage to choose the running parameters for your experiment.

Sequencing and analysis

**Start sequencing**
Configure and start a new sequencing run

**Start from a sample sheet**
Import sample sheet for sequencing

**Flow cell check**
Assess flow cell health

**Post sequencing analysis**
Analyse existing sequence data

**Custom scripts**
Start a run from a custom script

Utilities

**Hardware check**
Validate device is ready for sequencing

**Minimap2 converter**
Create a .mmi from a .fasta reference

Positions

2. Type in experiment name, select position, sample ID, and choose flow cell type from the drop down menu.
 - **Experiment Name:** DSCWorkshop2025
 - **Position:** X3
 - **SampleID:** DSC2025_RAD_Deplete
 - Flow cell ID automatically is filled in
 - Flow cell type should be FLO-MIN114

Positions

Experiment details

Experiment name *

DSCWorkshop2025

Device positions

MinION positions

GridION · INT-GRD-1577

[Deselect all positions](#)

▶

X1

X2

X3

X4

X5

Position	Flow cell ID	Flow cell type	Sample ID
X1	FAX56448	FLO-MIN114 ▾	DSC2025_RAD_Deplete

3. Select Continue to move to the Kit Selection.

4. Select the kit used from the Kit selection menu: **Rapid Sequencing Kit (SQK-RAD114)**

Kit selection

Sample type

DNA

RNA

PCR-free

PCR

PCR-free

Multiplexing

Yes

No

☐ Control

[Reset filters](#)

Ligation Sequencing Kit SQK-LSK114	Ligation Sequencing Kit SQK-LSK114-XL	Rapid Barcoding Kit 96 SQK-RBK114-96	Native Barcoding Sequencing Kit 24 SQK-NBD114-24
Native Barcoding Sequencing Kit 96 SQK-NBD114-96	Ultra-Long DNA Sequencing Kit SQK-ULK114	16S Barcoding Kit SQK-16S114-24	Multiplex Ligation Sequencing Kit XL SQK-MLK114-96-XL
PCR cDNA Barcoding Kit - 24 SQK-PCB114-24	PCR-cDNA Sequencing Kit SQK-PCS114	Rapid Sequencing Kit SQK-RAD114	Rapid Barcoding Kit 24 SQK-RBK114-24
Rapid PCR Barcoding Kit SQK-RPB114-24			

5. Select Continue to move to the Run configuration.

6. In Sequencing and analysis, basecalling, and Adaptive sampling, and advanced sequencing options can be modified.
 - a. Ensure Basecalling is selected (On)
 - b. Edit the model to select **Super-accurate basecalling** with Modified bases Off.
 - c. Ensure Barcoding is Off
 - d. Ensure Alignment is Off
 - e. Turn on Adaptive sampling
 - i. **Select Enrich**
 - ii. Ensure Basecall on-target reads only is Off
 - iii. **For Reference for alignment, choose the appropriate genome from the location of your reference genomes**
 - iv. Under advanced options, change the **End channel to 256**

Sequencing and analysis

Basecalling ☒

Model

Modified bases

Super-accurate – v4.3.0 – 400bps

Off

Barcoding ☐

Alignment ☐

Adaptive sampling ☒

Method

Basecall on-target reads only

Selected channels

Reference

Bed file

Deplete

Off

1 – 256

Bacillus_subtilis_complete_genome.fasta

No file selected

Advanced sequencing options

7. In Data targets, change the Run limit to Stop run when Time equals your desired run length (for the workshop we initially used 18 hrs).

Data targets

Conditions

Run limit

Stop run when sequencing reaches 18 hours

8. Output options can be left with defaults.
- a. File location: /data/
 - b. Data format: FASTQ
 - c. Output frequency: Every 10 minutes
 - d. FASTQ compression ON
 - e. Raw reads: POD5
 - f. Filtering On, Min Qscore = 10

Output

File output options

File location

/data/

346.2 GB used – 2.9 TB free

Data format

FASTQ

Output frequency

Every 10 minutes

FASTQ compression

On

Raw reads

POD5

Filtering

☒

Min Q score

10

9. Select 'Start' to run the experiment.

Barcoding - MinKnow

Commented [2]: @ljmullen2@alaska.edu please review
Assigned to ljmulen2@alaska.edu

Starting a Sequencing Setup on GridION

1. Select the 'Start Sequencing' option on the Start homepage to choose the running parameters for your experiment.

Sequencing and analysis

**Start sequencing**
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Start a run from a custom script

Utilities

**Hardware check**
Validate device is ready for sequencing

**Minimap2 converter**
Create a .mimi from a .fasta reference

Positions

2. Type in experiment name, select position, sample ID, and choose flow cell type from the drop down menu.
 - **Experiment Name:** DSCWorkshop2025
 - **Position:** X4
 - **SampleID:** DSC2025_RBK_Replicates
 - Flow cell ID automatically is filled in
 - Flow cell type should be FLO-MIN114

Positions

Experiment details

Experiment name *
DSCWorkshop2025

Device positions

MinION positions

GridION · INT-GRD-1577
[Deselect all positions](#)

▶

X1

X2

X3

X4

X5

Position	Flow cell ID	Flow cell type	Sample ID
X1	FAX56448	FLO-MIN114 ▾	DSC2025_RBK_Replicates

3. Select Continue to move to the Kit Selection.

4. Select the kit used from the Kit selection menu: **Rapid Sequencing Kit (SQK-RBK114-24)**

Kit selection

Sample type

DNA

RNA

PCR-free

PCR

PCR-free

Multiplexing

Yes

No

☐ Control

[Reset filters](#)

Ligation Sequencing Kit SQK-LSK114	Ligation Sequencing Kit SQK-LSK114-XL	Rapid Barcoding Kit 96 SQK-RBK114-96	Native Barcoding Sequencing Kit 24 SQK-NBD114-24
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PCR cDNA Barcoding Kit - 24 SQK-PCB114-24	PCR-cDNA Sequencing Kit SQK-PCS114	Rapid Sequencing Kit SQK-RAD114	Rapid Barcoding Kit 24 SQK-RBK114-24 ✓
Rapid PCR Barcoding Kit SQK-RPB114-24			

5. Select Continue to move to the Run configuration.

6. In Sequencing and analysis, basecalling, and Adaptive sampling, and advanced sequencing options can be modified.
- Ensure Basecalling is selected (On)
 - Edit the model to select **Super-accurate basecalling** with Modified bases Off.
 - Ensure Barcoding is On
 - Trim is Off
 - Selected barcodes All
 - Ensure Alignment is Off
 - Ensure Adaptive sampling is Off

Sequencing and analysis

Basecalling

Model

Super-accurate – v4.3.0 – 400bps

Modified bases

Off

Barcoding

Trim

Off

Selected barcodes

All

Alignment

Adaptive sampling

Advanced sequencing options

7. In Data targets, change the Run limit to Stop run when Time equals your desired run length (for the workshop we initially used 18 hrs).

Data targets

Conditions

Run limit

Stop run when sequencing reaches 18 hours

8. Output options can be left with defaults.
- a. File location: /data/
 - b. Data format: FASTQ
 - c. Output frequency: Every 10 minutes
 - d. FASTQ compression ON
 - e. Raw reads: POD5
 - f. Filtering On, Min Qscore = 10

Output

File output options

File location

/data/

346.2 GB used – 2.9 TB free

Data format

FASTQ

Output frequency

Every 10 minutes

FASTQ compression

On

Raw reads

POD5

Filtering

☒

Min Q score

10

9. Select 'Start' to run the experiment.

Ligation - MinKnow

Commented [3]: @alehr2@alaska.edu please review
Assigned to alehr2@alaska.edu

Starting a Sequencing Setup on GridION

1. Select the 'Start Sequencing' option on the Start homepage to choose the running parameters for your experiment.

Sequencing and analysis

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Utilities

**Hardware check**
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**Minimap2 converter**
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Positions

2. Type in experiment name, select position, sample ID and choose flow cell type from the drop down menu.
- **Experiment Name:** DSCWorkshop2025
 - **Position:** X5
 - **SampleID:** DSC2025_LSK_Yield
 - Flow cell ID automatically is filled in
 - Flow cell type should be FLO-MIN114

Positions

Experiment details

Experiment name *
DSCWorkshop2025

Device positions

MinION positions

GridION · INT-GRD-1577
[Deselect all positions](#)

X1

X2

X3

X4

X5

Position	Flow cell ID	Flow cell type	Sample ID
X1	FAX56448	FLO-MIN114 ▾	DSC2025_LSK_Yield

3. Select Continue to move to the Kit Selection.

4. Select the kit used from the Kit selection menu: **Rapid Sequencing Kit (SQK-LSK114)**

Kit selection

Sample type

DNA

RNA

PCR-free

PCR

PCR-free

Multiplexing

Yes

No

☐ Control

[Reset filters](#)

Ligation Sequencing Kit SQK-LSK114	Ligation Sequencing Kit SQK-LSK114-XL	Rapid Barcoding Kit 96 SQK-RBK114-96	Native Barcoding Sequencing Kit 24 SQK-NBD114-24
Native Barcoding Sequencing Kit 96 SQK-NBD114-96	Ultra-Long DNA Sequencing Kit SQK-ULK114	16S Barcoding Kit SQK-16S114-24	Multiplex Ligation Sequencing Kit XL SQK-MLK114-96-XL
PCR cDNA Barcoding Kit - 24 SQK-PCB114-24	PCR-cDNA Sequencing Kit SQK-PCS114	Rapid Sequencing Kit SQK-RAD114	Rapid Barcoding Kit 24 SQK-RBK114-24
Rapid PCR Barcoding Kit SQK-RPB114-24			

Select expansion pack

Cas9 Sequencing Expansion EXP-C9S001	PCR Barcoding Expansion 1-12 EXP-PBC001	PCR Barcoding Expansion 1-96 EXP-PBC096	
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 - b. Edit the model to select **Super-accurate basecalling** with Modified bases Off.
 - c. Ensure Barcoding is Off
 - d. Ensure Alignment is Off
 - e. Ensure Adaptive sampling is Off

Sequencing and analysis

Basecalling

Model

Super-accurate – v4.3.0 – 400bps

Modified bases

Off

Barcoding

Alignment

Adaptive sampling

Advanced sequencing options

7. In Data targets, change the Run limit to Stop run when Time equals your desired run length (for the workshop we initially used 18 hrs).

Data targets

Conditions

Run limit

Stop run when sequencing reaches 18 hours

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Output

File output options

File location

/data/

346.2 GB used – 2.9 TB free

Data format

FASTQ

Output frequency

Every 10 minutes

FASTQ compression

On

Raw reads

POD5

Filtering

☒

Min Q score

10

9. Select 'Start' to run the experiment.